## Contact

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www.linkedin.com/in/dhardestylewis (LinkedIn)
github.com/dhardestylewis (Portfolio)
www.tacc.utexas.edu/about/directory/daniel-lewis (Company)


## Top Skills

Geographic Information Systems (GIS)
MATLAB
R (Programming Language)

## Languages

Spanish (Professional Working)
French (Limited Working)
English (Native or Bilingual)

## Honors-Awards

Fellowship

## Publications

Artificial Intelligence for Modeling Complex Systems

# Daniel Hardesty Lewis


Engineering scientist associate at Texas Advanced Computing Center (TACC)
Austin, Texas, United States


## Summary


I have deep experience applying high-performance computing techniques to terrain and flood models, developing reproducible and scalable workflows from source data all the way out to web services, at scales as small as a parcel and as large as countries. I am currently developing the next generation of highest quality terrain data. I have a broad base of experience in research engineering, front- and back-end web development, and data analysis. Some of the reproducible water models that I have developed have more than 50,000 downloads from the hydro community. I am also highly active in my local community, serving as a catalyst for positive change on behalf of and impacting more than 10,000 residents.


## Experience

### Texas Advanced Computing Center (TACC)

5 years 10 months

#### Engineering scientist associate

August 2021 - Present (2 years 4 months)
Austin, Texas, United States

I design and implement intelligent decision support systems to support computational science, disaster mitigation and response, and resiliency research projects.

principal and sole developer of next generation highest quality seamless terrain map of Texas
scale hydro models such as HAND & HEC-RAS on some of the most powerful computers in the world
develop SOWs between UT, A&M, and TNRIS/USACE
prepare computing facility for rapid response to drought, hurricane, flood disasters with replicable scaling of engineering workflows and training

#### Research engineering / scientist associate I

February 2018 - July 2021 (3 years 6 months)

Austin, Texas, United States

I contribute research towards integrated modelling and water resource management within the state of Texas and abroad, and I've applied machine and deep learning approaches to problems for the energy industry.

### The University of Texas at Austin

Co-instructor
August 2020 - December 2020 (5 months)
Austin, Texas, United States

Helped teach Scientific Computation

### ICC Austin

6 years

System Architect
September 2014 - August 2020 (6 years)
Austin, Texas, United States

Re-designed and implemented entirely new internet infrastructures for 9 buildings serving 115 students, enabling gigabit speeds at each. Re-positioned new enterprise-level wireless access points. Trained organizational capacity to manage new infrastructure.

Conflict mediator
June 2015 - August 2018 (3 years 3 months)
Austin, Texas

Mediated disputes between members and their housing coops. Familiar with all applicable policies and bylaws. Guided facilitations which reached mutually agreed-upon solutions, to help ensure the continued success of this tenant self-managed housing organization.

Technology Officer
September 2014 - December 2017 (3 years 4 months)
Austin, Texas area

Set up and managed public computers for a 20+ tenant building.

### ICPSR at University of Michigan Institute for Social Research

Program Scholar
June 2019 - July 2019 (2 months)
Ann Arbor, Michigan, United States

The ICPSR Summer Program provides rigorous, hands-on training in statistical techniques, research methodologies, and data analysis. ICPSR Summer Program courses emphasize the integration of methodological strategies with

the theoretical and practical concerns that arise in research on substantive issues.

Courses enrolled:
- Intro to Network Analysis
- Machine Learning: Applications and Opportunities in the Social Sciences
- Math for Social Science III

### The University of Texas at Austin
Co-instructor
January 2018 - May 2018 (5 months)
Austin, Texas, United States

Helped teach Data Informatics and Intelligent Systems for Geosciences

### Texas Advanced Computing Center (TACC)
Undergraduate Research Assistant
May 2017 - August 2017 (4 months)
Austin, Texas, United States

With a $4000 fellowship from the Texas Institute for Discovery Education in Science, I helped update Texas's groundwater availability models to a more contemporary format

### MICEFA
Information Technology Intern
October 2016 - January 2017 (4 months)
Paris, Île-de-France, France

Developed Bash workflow to transform data from a proprietary database into SQL

### The University of Texas at Austin
Undergraduate Research Assistant
May 2013 - December 2013 (8 months)
Austin, Texas Area

The boundaries of the stellar evolution code MESA were tested, especially with regard to the high-mass, oxygen/neon-core white dwarf regime. Theoretical models were run, and the existence (by the standard physics as implemented in the program) of such a regime was found from these calculations.

## Education

The University of Texas at Austin

Bachelor of Science - BS, Mathematics · (2012 - 2017)

The University of Texas at Austin

Certificate, Scientific Computation